

Engineering principles explain biological systems better than evolutionary theory

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■ Introduction

Since antiquity, scholars have debated whether the appearance of design in life was the product of a transcendent mind or undirected natural processes. The latter view rose to dominance after the scientific community largely embraced Darwin's theory of natural selection. In the last few decades, the hegemony of the standard model for evolution has started to wane. Recent discoveries have forced biologists to replace evolutionary assumptions with design-based assumptions, language and methods of investigation. This trend is to a large extent driven by the observation that the same engineering motifs and patterns employed in human creations are pervasive in living systems. What is becoming increasingly clear is that engineering principles far better explain nearly every aspect of life than evolutionary theory. This conclusion

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perfectly coincides with the central Christian doctrine that life was designed by God and not an unintended accident of nature.

Since ancient times, scholars have debated whether everything in the world was simply the product of natural processes or derived from the plans of a supreme intellect (Lloyd 1970). The ancient form of the former philosophy is termed atomism. Its adherents believed that matter was composed of indivisible 'atoms' that interact according to various rules. The interactions of the atoms, chance and time resulted in everything observed in our world. The modern form of this philosophical tradition is termed scientific materialism.

The atomists developed evolutionary theories to explain the appearance of design in life. These theories seem eerily familiar to modern ears. The Greek physician Hippocrates proposed in the late 5th or early 4th century BC a model for heredity and adaptation that Charles Darwin described as nearly identical to his own (Tsiompanou & Marketos 2013). And the poet Lucretius developed in the 1st century BC an evolutionary framework based on a primitive form of natural selection (Campbell 2004). Lucretius' agenda was to remove teleology (aka design) and by extension the influence of any deity from the world.

In stark contrast, philosophers such as Plato and Aristotle rejected the claim that matter had in itself the capacity to self-organise into living creatures. Instead, they believed that a transcendent mind envisioned the architectural plan or design logic of every distinct creature. And the design plan detailed how the matter would be arranged in a creature's final form. That form would never have occurred without external direction.

Some scholars have argued that Aristotle's views are more compatible with scientific materialism than ID (Feser 2019). But such claims result from a misreading of Aristotle's writings (Bos 2003, 2018; Gerson 2005; Henry 2019). He, like Plato, would have aligned himself with modern design proponents far more than with scientific materialists.

The arguments the ancient philosophers employed to defend their conclusion of design in life (Archer-Hind 1888; Aristotle 2010) resemble those promoted by proponents of design today. And the atomists' justification for ignoring the evidence for design resembles arguments by modern scientific materialists (Sedley 2008). This conflict was reflected in the opening chapter of the Apostle Paul's letter to the Romans where he described how the clear evidence for God's 'eternal power' is 'understood by the things that are made' (Keener 2009). The apostle would have seen the design-evolution debate today as a continuation of the same conflict that raged in the 1st century.

Darwin developed the modern version of the evolutionary theory, and he, like his ancient predecessors, argued that evolutionary processes could mimic

the creative power of an intelligent agent in generating biological features of any level of complexity and ingenuity. More specifically, natural selection could act as a designer substitute empowering the environment to act as a creative agent, thus removing the need for a creator. Such reasoning allowed evolutionists to simply dismiss any evidence for design, no matter how conclusive, as merely an artifact of a blind, undirected process that did not have any end goal in mind.

Francisco Ayala, a former president of the American Association for the Advancement of Science, described Darwin's achievement of returning the atomist philosophical tradition to dominance (Ayala 2007) in the clearest of possible terms:

Darwin's greatest contribution to science is that he completed the Copernican Revolution by drawing out for biology the notion of nature as a system of matter in motion governed by natural laws. With Darwin's discovery of natural selection, the origin and adaptations of organisms were brought into the realm of science. The adaptive features of organisms could now be explained, like the phenomena of the inanimate world, as the result of natural processes, without recourse to an intelligent designer. (p. 8567)

Over the next century, biologists combined Darwin's theory with genetics and population genetics to form neo-Darwinism, also known as the Modern Synthesis. In the following decades, the theory was widely accepted to explain away all the evidence for design in life as mere illusion. This belief, however, is becoming increasingly untenable in light of developments in the biological sciences over the past several decades.

■ The demise of neo-Darwinism

To name but a few lines of evidence, the fossil record demonstrates that novel organisms consistently appear suddenly. Approximate ancestor-descendent series of intermediate forms that connect any organism to a common ancestor with a distinctly different species have never been identified (Meyer 2013). And no species has ever been observed to substantially change during its tenure on earth.

Palaeontologists often point to 'transitional fossils' to argue that the record supports evolutionary narratives. An iconic example is the purported whale series that includes a sequence of fossils 'transitioning' from a terrestrial animal to one that is fully aquatic. What is often omitted is that such sequences do not correspond to anything close to an ancestor-descendent series. The creatures are at best cousins that reside at the ends of their own separate branches in the hypothetical tree of life (TOL) (Wells 2017). None has conclusively been demonstrated to be ancestral to or a descendent of the others. They each appear suddenly, and they do not appreciably change after their appearance.

The species share some common features with each other, but equally striking similarities exist between organisms that are not believed to be closely related. Given the millions of species that have existed, some set will inevitably appear to line up in a sequence purely by chance. One can find numerous comparable sequences in human creations such as the following: unicycle → bicycle → motorcycle → convertible car → station wagon → bus. A motorcycle could be described as a transition between a bicycle and a car, but that descriptor does not mean that the three share a common ancestor that incrementally evolved into the modern transportation vehicles. The same holds true with so-called transitional fossils.

Leading evolutionary theorist Eugene Koonin acknowledges the fact that the evolutionary processes seen today cannot explain the sudden appearances of new body plans – a body plan is the distinctive design architecture or blueprint shared by a group of organisms such as fish. He compares the sudden appearance of a new body plan to the cosmological Big Bang (Koonin 2007). He proposes that large-scale transformations (e.g. fish to amphibians) could have been driven by ‘various processes of genetic information exchange, such as horizontal gene transfer, recombination, fusion, fission and spread of mobile elements’. But this assertion contradicts all observational and experimental evidence of what such processes could accomplish.

Major transformations require alterations to an organism’s underlying architecture, and such changes must result from modifying the early stages of a fertilised egg developing into an embryo and an embryo developing into an offspring (or the equivalent). The challenge is that all alterations early in development that generate significant modifications are harmful. Leading developmental biologist Eric Davidson described how alterations to the developmental gene regulatory networks⁵⁵ underlying the embryological construction of body plans are always ‘catastrophically bad’ (Davidson 2011), implying that complex organisms can never significantly change because of any undirected process.

A related challenge is the implausibility of many novel proteins evolving by chance. Proteins are chains consisting of primarily 20 possible amino acids. The proper sequences will result in the chain folding into a complex three-dimensional structure that can serve as a structural component of a cell or act as an enzyme that drives a specific chemical reaction. The barrier for evolving novel proteins is the fact that the percentage of sequences that fold into functional enzymes is infinitesimally small. Protein evolution expert Doug Axe estimated the proportion of sequences that folded into one portion of the β -lactamase enzyme at 1 in 10^{77} – a 1 followed by 77 zeros (Axe 2004).

55. A developmental gene regulatory network is a set of genes that coordinate to direct the development of an egg into an embryo and then into an offspring or the equivalent process.

Later research showed that the actual rarity is far more extreme for that protein and for many others.⁵⁶ Such small targets in sequence space (graph of all possible amino acid sequences) could never be located by an undirected search, so many proteins could not have originated through an evolutionary process.

Leading expert Dan Tawfik acknowledged that the origin of novel proteins is a complete mystery (Mukhopadhyay 2013):

Once you have identified an enzyme that has some weak, promiscuous activity for your target reaction, it's fairly clear that, if you have mutations at random, you can select and improve this activity by several orders of magnitude [...] What we lack is a hypothesis for the earlier stages, where you do not have this spectrum of enzymatic activities, active sites and folds from which selection can identify starting points. Evolution has this catch-22: Nothing evolves unless it already exists. (n.p.)

He even described it as 'something like close to a miracle'.

Compounding the problem, the maximum possible time for major transitions (e.g. a few million years for the appearance of a new phylum) is sufficient for only the tiniest fraction of the required new genetic information to appear and spread throughout a population. For instance, the entire purported evolutionary history between an ape-like ancestor of humans to modern humans is insufficient for even two coordinated mutations to appear in any individual (Durrett & Schmidt 2008). Yet even simple evolutionary steps require far more specific mutations, such as the origin of a targeted mid-range neural connection. And time scales grow exponentially with the required number of specific mutations (Hössjer, Bechly & Gauger 2021). The disparity between the time available for major transitions and the minimal required time to generate the new genetic information is comparable to the difference between the world-record pole vault height and the distance to Alpha Centauri.

Such problems compelled leading evolutionary theorists to organise the Royal Society meeting titled *New Trends in Evolutionary Theory* in 2016. Speakers acknowledged the fact that the standard evolutionary model cannot adequately address the aforementioned challenges or related issues. In particular, natural selection was recognised as having no true creative power.

56. Researchers determined for β -lactamase (Bershtein et al. 2006) and HisA (Lundin et al. 2018) the equations for the percentage of operational proteins as a function of the number of alterations to their amino acid sequences. The function declines hyper-exponentially with increasing numbers of alterations. The total number of functional sequences can be estimated by multiplying the fitness function by the number of sequences with n alterations away from the original sequence and then summing for all n . The percentage of functional sequences (aka rarity) equates to the number of functional sequences divided by the total number of sequences of length L , the protein's sequence length. The resulting percentage of functional sequences is much smaller than Axe's estimate. Axe looked at the rarity in a region of sequence space close to a functional sequence. The actual rarity is much more extreme when all of sequence space is considered. Moreover, β -lactamase is an enzyme that performs a relatively simple function, so a significant percentage of proteins almost certainly correspond to more extreme rarity.

Yet not one presentation offered concrete evidence that any of the proposed extensions to evolutionary theory could fill the standard model's explanatory deficits (Why the Royal Society Meeting Mattered, in a Nutshell 2016).

With natural selection off the table as a designer substitute, materialist biologists are left in a situation comparable to the crisis astronomy would face if physicists discovered that gravity stopped operating beyond 10 000 kilometres past a celestial body. The loss of the only feasible mechanism that could explain the motion of planets, stars and galaxies would result in absolute pandemonium and despair. Most biologists will not come to terms with their true predicament so easily as evolution operates not only as a scientific theory but as a sacrosanct creation narrative for a secular society.

■ The erosion of scientific materialism

Despite the reticence of many to question the official scientific orthodoxy, a scientific revolution has begun in a biological subdiscipline known as systems biology (Miller 2021a). Biologists and engineers have come together to better understand the higher-level organisation of living systems. Their discoveries have forced them to replace historic evolutionary assumptions with design-based assumptions, language and methods of investigation.

This transformation in thinking reflects how the philosophical foundation of scientific materialism, which has defined science for the past two centuries, is eroding in the face of the most recent biological data. The traditional approaches implemented in biological research were founded on reductionism – the belief that studying the physical and chemical interactions between biological molecules should eventually lead to an understanding of life's higher-level operations and organisation.

This assumption was central to evolutionary thinking as evolutionary processes can differentiate the results only from individual changes to DNA or other inherited information-bearing structures (Wells 2014). Such changes modify the lowest-level chemical interactions or structural details. Nearly every change must immediately benefit the organism, or it would not typically persist in the population. Consequently, the function of a biological system should represent the sum of the functions of each part. An analogy would be a team of cleaners whose net benefit is simply the sum of the activities of each individual cleaner.

Based on this framework, biologists expected that studying the actions of individual reactions and structural molecules would allow them to discern the operations of larger systems and structures. The role of the larger system should represent a summation of the roles of each individual part. Removing one part should simply reduce the function of the system by the minor role the removed part played.

More commonly today, systems biologists reject this reductionist approach as it has failed to yield any significant understanding of the organisation of living systems. Instead, they now recognise that they must look at life as a collection of integrated systems composed of integrated components where the whole is greater than the sum of the parts (also known as holism). In other words, Michael Behe's concept of irreducible complexity (Behe 2020) has implicitly become a central tenant of the field. Researchers would rarely use Behe's exact language or recognise the implications, but they are tacitly acknowledging the ubiquity of this feature in life.

Philosopher of science Michel Morange (2008) describes critiques of the traditional approach to biology in his provocatively titled article 'The Death of Molecular Biology?':

[D]oes it mean that molecular biology is dead, and has been displaced by new emerging disciplines such as systems biology and synthetic biology? Maybe its reductionist approach to living phenomena has been substituted by one that is more holistic [...] Some even consider the age of molecular biology as having been a period of extreme misorientation of biological research, an error that it is high time to repair. (p. 31)

Similarly, Virologist Derek Gatherer (2010) comments:

The epistemological antireductionist is a holist because complete reductionism is technically impossible. For a systems biologist, sheer complexity, one might say irreducible complexity, can prevent the exhaustive analysis of a network of any interesting size [...]. The broadening of molecular biology into systems biology has created a situation where researchers have a vague inkling that their underlying philosophy is in need of refurbishment, and holism appears to offer much of what is wanted. (p. 4)

It is worth noting that if a system is too 'irreducibly complex' to even describe, it is certainly too specified and complex to originate through any undirected process (Ewert, Marks & Dembski 2013).

■ The return of teleology

Part and parcel with the erosion of the reductionist assumption has been the implicit acceptance of the assumption that teleology is central to life (Miller 2021g). Teleology is the philosophical term for design or purpose. This change is cataclysmic as the gatekeepers at most mainstream scientific institutions have historically forbidden researchers from explicitly appealing to teleology as an explanation.

Despite this prohibition, biologists have intuitively recognised that identifying the purpose of a structure or system was essential in understanding it. Yet they would not explicitly acknowledge their reliance on design-based reasoning. The German physician and physiologist Ernst Wilhelm Ritter von Brücke best described this tension (Krebs 1954, p. 45) 'teleology is a lady

without whom no biologist can live. Yet he is ashamed to show himself with her in public.' The difference now is that the appeal to teleology is ubiquitous and explicit.

The scientific elite has attempted to explain the appearance of teleology as a by-product of evolutionary mechanisms. But the most philosophically minded biologists have come to recognise that appealing to such undirected processes as a designer agent is logically incoherent. Biologist David Hanke emphasises this point in his essay, 'Teleology: the explanation that bedevils biology' (Hanke 2004):

Biology is sick. Fundamentally unscientific modes of thought are increasingly accepted, and dominate the way the subject is explained to the next generation. The heart of the problem is that we persist in making (literal) sense of the world that we now know to be senseless by attributing subjective values to the objects in it, values that have no basis in reality. (p. 143)

Hanke (2004) lists several statements by none other than atheist Richard Dawkins that embody this philosophical transgression:

Somatic cell divisions are used to make mortal tissues, organs and instruments whose 'purpose' is the promoting of germ line divisions.

The replicators that exist tend to be the ones that are good at manipulating the world to their own advantage. (pp. 143-144)

Hanke (2004) acknowledges the fact that life might display clear signatures of design to those uninitiated into scientific materialism, but he insists that reality must be the opposite of where the evidence naturally leads:

The bits of living things at all levels of scale from molecules to the whale's tail also happen to have symmetry, precision, and complexity, clues that simply shout 'purpose' in the inquiring mind. This has to be wrong. Because they are not manufactured they cannot have been designed, and so no one ever had a purpose for them. They make themselves and so just exist, without purpose or intended use. (pp. 145-146)

He dedicates the rest of his essay to arguing that attempts to identify the intent of a biological structure or trait lead biologists astray. Investigators should instead assume that their objects of study simply exist without any higher purpose to meet some biological goal.

Zoologist John Reiss (2009) presents similar denunciations in his book *Not by Design: Retiring Darwin's Watchmaker*. He aims to convince biologists to purge all teleological language from their writing and thinking:

Life is not designed, or at least it shows no evidence of design for anything other than continued existence, which needs no design [...] To truly retire the watchmaker, [...] We must admit that there is not only not design but indeed not even 'apparent design' in the biological world, in the sense of entities doing any more than they need to do to continue to exist. (p. 356)

He proposes that biologists instead appeal to palaeontologist Georges Cuvier's concept of the 'conditions of existence' in a manner that appears to be the biological equivalent of the weak anthropic principle in cosmology (Meyer 2021).

Hanke and Reiss's attempts to expel teleological language from biology in the face of the torrent of opposing data have proven as effective as attempting to use a toy sand bucket to hold back a striking tsunami. The most astute biologists now recognise that the only feasible approach to understanding biological systems is to understand their purpose. This realisation appears explicitly in the book *System modeling in cellular biology* (SMCB) (Szallasi, Stelling & Periwal 2007):

A hope for understanding complexity in biology then is to uncover operational principles through a 'calculus of purpose' (Lander 2004) – by asking teleological questions such as why cellular networks are organized as observed, given their known or assumed function. (p. 24)

Other philosophers of science have also expressed the need to apply 'design explanations' to biology (Brillard 2010). Philosophers and complexity theorists Mantri and Thomas explained both the need for and the resistance against this change (Mantri and Thomas 2019):

Unfortunately, research in the world of modern biology is currently divorced from that of design-theory. Yet each discipline could benefit from studying the other. From a design perspective (and subject to environment/precedent constraints), form seems to be following function (e.g. the elbow joint of the fore-arm for bringing food to the mouth). The fundamental problem associated with design in biology, is that of agency. [...] In this paper, we try to bridge the seemingly insurmountable gap between design-theory and biological 'designs,' without getting derailed by 'intelligent design' polemics. (p 1)

Mantri and Thomas desperately attempt to reframe 'design-theory' within the confines of evolutionary assumptions, but their efforts amount to little more than invoking phrases such as 'stigmergic teleology' and 'emergence' without providing any substantive details of what such concepts would look like in an actual evolutionary scenario. They simply maintain that life looks like a duck, swims like a duck and quacks like a duck, but it is actually a cat.

■ The assumption of optimality

A second changing assumption in systems biology relates to the optimality of biological systems (Miller 2021h). The underlying logic of evolutionary models predicts that suboptimal design and non-functional remnants of organisms' evolutionary past should litter the biosphere. Philosopher Philip Kitcher emphasised this point in his book *Living with Darwin: Evolution, Design, and the Future of Faith*. He appealed to examples of what he believed were clumsy,

incompetent designs and non-functional DNA regions as his primary argument for dismissing ID and accepting evolution (Kitcher 2007):

If you were a talented engineer designing a whale from scratch, you probably wouldn't think of equipping it with a rudimentary pelvis [...] If you were designing a human body, you could surely improve on the knee. And if you were designing the genomes of organisms, you would certainly not fill them up with junk. (p. 57)

In a similar vein, biologist Nathan Lents argued in his book *Human Errors: A Panorama of Our Glitches, from Pointless Bones to Broken Genes* that the 'bungling' design seen throughout the human body demonstrates that we are not the product of an intelligent designer but an undirected evolutionary process (Lents 2018):

The third category features those human defects that are due to nothing more than the limits of evolution. All species are stuck with the bodies that they have and they can advance only through the tiniest changes, which occur randomly and rarely. We inherited structures that are horrendously inefficient but impossible to change [...].

This is why our throats convey both food and air through the same tiny space and why our ankles have seven pointless bones sloshing around. Fixing either of those poor designs would require much more than one-at-a-time mutations could ever accomplish. To suppose that these living things were separately created is to view the creative agent as whimsical, bungling, a mediocre engineer, an unintelligent designer. (pp. xi-xii)

The expectation of poor design is not simply a subjective conclusion based on intuition, but it has been rigorously demonstrated in computational models. One such model created by Snoke, Cox and Petcher (2015) elucidated why evolutionary processes that allow for increases in complexity must generate large quantities of junk DNA and non-functional elements. The details of their model are complex, but the underlying logic is straightforward.

For complex innovations to emerge, organisms must allow non-functional DNA to appear and persist in the population until a functional sequence arises. Such additions to the genome could occur through a gene duplicating and then repeatedly mutating. Junk DNA would inevitably accumulate to encompass a significant percentage of the genome. This requirement is why biologists once assumed that junk DNA comprised as much as 97% of the human genome (Hall 2012). Similarly, the origin of complex structures (e.g. molecular machines) requires countless trial-and-error arrangements of molecules or tissues until something advantageous appears. Most of the trials would be either non-functional or inefficient. Consequently, only a minority of biological structures and systems should appear highly optimised.

In stark contrast to evolutionary predictions, mounting evidence demonstrates that life consistently demonstrates optimal design. Most of the examples of allegedly poor design cited by Kitcher, Lents and other sceptics have been overturned such as the human appendix (Dunn 2012), wisdom teeth

(Lieberman 2013), eyes (Wells 2017) and spine (Nordin 2021). The remaining ones typically represent degradations of once optimal designs or appeal to the imperfection-of-the-gaps fallacy.

Purported examples of poor design usually represent opinions resulting from armchair critics' limited understanding of the technical literature and their lack of training in engineering. For instance, in direct contradiction to Kitcher's and Lents's assertions, engineers commonly reuse design motifs in new ways, just as is seen with the whale pelvis. Medical professionals and engineers have demonstrated how the human knee and ankle are optimally and exquisitely designed (Günther et al. 2004; Seo & Yi 2009). Engineers have even looked to these structures for inspiration in designing artificial limbs (Burgess & Etoundi 2014; Etoundi, Burgess & Vaidyanathan 2013).

Moreover, most of the human genome is now known to be functional because of the Encyclopedia of DNA Elements (ENCODE) project (Abascal et al. 2020). The devastating ramifications of this revelation for evolutionary theory have not gone entirely unnoticed. Biochemist Dan Graur bluntly stated (Graur 2013):

If the human genome is indeed devoid of junk DNA as implied by the ENCODE project, then a long, undirected evolutionary process cannot explain the human genome. If, on the other hand, organisms are designed, then all DNA, or as much as possible, is expected to exhibit function. If ENCODE is right, then evolution is wrong. (slide 16)

Of equal significance, systems biologists increasingly recognise that assuming optimal design leads to the most productive research. For instance, Nikolaos Tsiantis, Eva Balsa-Canto and Julio R. Banga developed in their 2018 Bioinformatics article (Tsiantis, Balsa-Canto & Banga 2018) a model for studying biological systems based on identifying 'underlying optimality principles'. Tsiantis, Balsa-Canto and Banga (2018) also surveyed leading researchers who also demonstrated the predictive power of assuming optimality:

Sutherland (2005) claims that these optimality principles allow biology to move from merely explaining patterns or mechanisms to being able to make predictions from first principles. Bialek (2017) makes the important point that optimality hypotheses should not be adopted because of esthetic reasons, but as an approach that can be directly tested through quantitative experiments. Mathematical optimization could therefore be regarded as a fundamental research tool in bioinformatics and computational systems biology. (pp. 2433-2434)

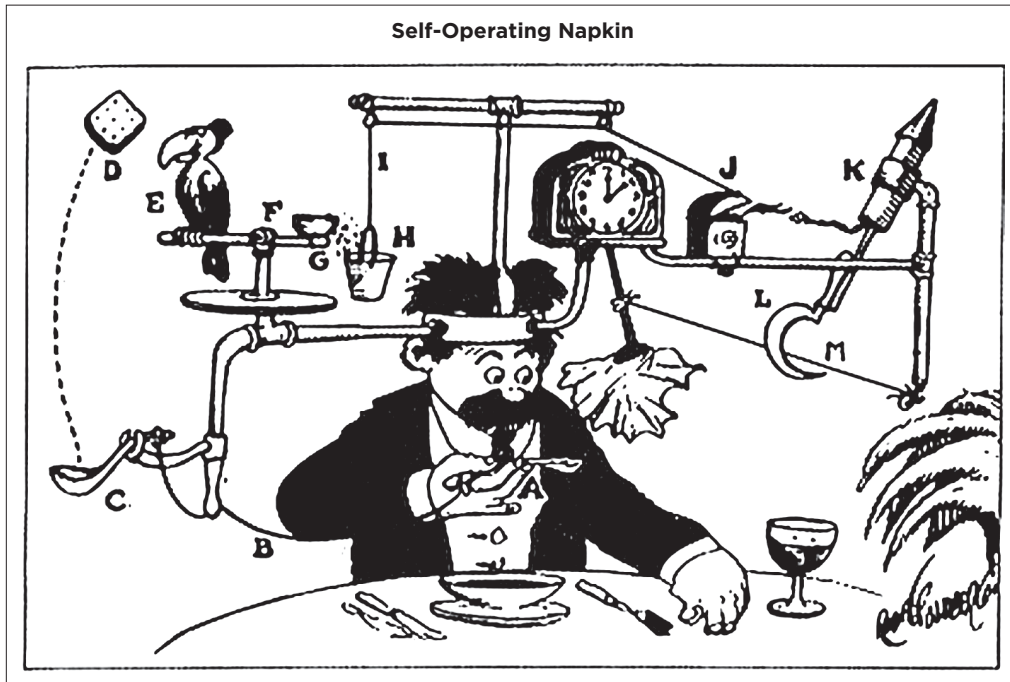
Other investigators have even shown that many biological systems operate at efficiencies close to the limits of what is physically possible. Examples include DNA replication and translation (Piñeros & Tlustý 2019), embryological development (Petkova et al. 2019) and sensory processes (Field & Sampath 2017). Human engineering pales in comparison to such achievements.

■ Human engineering and biology

A third changing assumption relates to the connection between human engineering and biological designs (Miller 2021c). Evolutionary theory predicts that biology should resemble human engineering only marginally at best. The underlying logic dictates that the components of complex biological structures and traits came together haphazardly without the benefit of foresight or goal-direction by an intelligent agent. Entities (e.g. proteins) that served one purpose linked with other entities serendipitously to achieve a different collective outcome (e.g. ATP synthase recharging ATP). The lowest-level pieces at the bottom of the organisational hierarchy originated first, so the final product represents a 'bottom-up' design.

In reality, any successful bottom-up design still requires designing the components and properly linking them together, but I will overlook this inconvenience for evolutionary theory for the sake of discussion.

Such composite systems should resemble Rube Goldberg machines (Figure 7.1). Rube Goldberg was an American inventor and cartoonist who drew contraptions that performed a simple task through a series of awkwardly interconnected unrelated devices. The action of one device would trigger the next, which would trigger the next and so on, to achieve some goal not directly



Source: Rube Goldberg, https://en.wikipedia.org/wiki/Rube_Goldberg_machine, published under public domain permissions as allowed by Wikimedia Commons.

FIGURE 7.1: 'Professor Butts and the Self-Operating Napkin' (1931).

connected to any of the individual components. The entire mechanism was comically inefficient and cumbersome.

Evolutionist Jerry Coyne has repeatedly emphasised how life should display a similar bottom-up design (Coyne 2019):

Indeed, the uniform experience of scientists who work on these systems is that they embody an absurd, Rube Goldberg-like complexity that makes no sense as the handiwork of an engineer but makes perfect sense as a product of a long and unguided historical process. (n.p.)

His insight into what evolution should produce is entirely correct. But he has yet to identify a single example of Rube Goldberg-like complexity in life because none exists.

In direct opposition to Coyne, systems biologists now recognise the fact that biology demonstrates a top-down design where an overarching goal and corresponding design constraints dictate the engineering of a complex trait. Each component of a structure or system perfectly integrates with other members to achieve a predetermined goal with astonishing efficiency. Thus, life does not resemble Rube Goldberg-type machines but human engineering. This stark contrast between evolutionary expectations and the actual high-level organisation of life has created tension within the biological community. The authors of SMCB commented (Szallasi, Stelling & Periwal 2007):

An often noted reservation against the type of analogies between biological and engineered systems we brought forward states that these two types of complex systems arise in fundamentally different ways, namely through evolution versus purpose-driven, top-down design (see e.g. Bosl & Li 2005). (p. 32)

Biology does not simply resemble human engineering generically, but it contains the very same design frameworks. Szallasi, Stelling and Periwal (2007) explain:

At a more abstract level, we see highly organized and structured networks that facilitate global and coordinated responses to variations in the environment on all time scales, using local and decentralized mechanisms [...] The basic framework is employed in many advanced technological systems [...] Clearly, from an engineering point of view, biology is a marvel of technological 'design'. We argue that analogies with engineered systems, in particular regarding how to generate appropriate responses to variations, are one major requirement on all highly integrated systems that can help us grasp biological complexity. (pp. 26–27)

The authors further argue that design motifs employed in life are known to represent the most effective strategies for achieving target goals (Szallasi, Stelling & Periwal 2007):

From engineering, it is known that feedback control (plus feedforward control) enabled by fast and if possible remote advanced-warning sensing is the most powerful mechanism for providing robustness to fluctuations in the environment and the component parts. The heat-shock response in *E. coli* appears to employ exactly the same principles as shown by detailed modeling and subsequent model reduction to the core elements (El-Samad et al. 2005). (p. 39)

Equally striking, Gregory Reeves and Curtis Hrischuk identify a cell as an embedded computing system (Reeves & Hrischuk 2016) as human and biological computing systems share numerous components. Examples include processing engine, information code, primary memory, secondary memory, memory addressing, low-level memory layout, memory management, cache, timer, randomly accessed persistent storage, high-level data formatting and the list goes on. These components also share many of the same functional interrelationships. Computer engineers would attest that such systems can only operate if all essential elements exist and are properly interconnected.

The most philosophically astute materialist scientists recognise the hazard of comparing biology to engineering too closely. Not only does engineering embody ID, but engineers have developed a deep intuition of what incremental processes can and cannot achieve. And they recognise that the design patterns pervasive in life could not possibly have emerged through any gradual, undirected process.

In response, biologists wedded to scientific materialism have argued that life is so different from human artifacts that they can dismiss engineers' conclusions about organisms' limited evolvability. A central fallacy in this argument is that they underestimate the fundamental similarities. Design motifs such as four-bar linkages and control systems must meet exacting requirements whether implemented in a space shuttle or a fish. Many of these requirements operate largely independently of the constituent materials that compose them or the exact methods they employ in their operation.

Moreover, the distinctive nature of living systems entails many additional requirements and even stricter constraints. For instance, the vertebrate eye must conform to many of the same or comparable requirements seen in digital cameras (Keum et al. 2018). And its construction requires a highly coordinated manufacturing process in embryology directed by a meticulously engineered genetic control system (Ogino et al. 2012). Not only must a biological element function properly, but an organism must also manufacture, maintain and operate it. These additional requirements translate into insurmountable barriers for any evolutionary scenario (Miller 2017a, 2017b).

In addition, the fundamental differences between human and biological engineering reflect the superiority of the latter. For instance, an international team of biologists and chemists detailed the extraordinary ingenuity associated with the materials that organisms manufacture (Frølich et al. 2017). And nanotechnology engineer Michael Simpson and his team identified how the cells' complexity and capabilities far exceed those of human technology (Simpson et al. 2004):

The genetic and biochemical processes that generate the complex and versatile behavior of cells operate as highly functional, densely packed, information-processing systems. [...] At the heart of this functionality are complex genetic

regulatory circuits and networks that process information in a manner similar to engineered circuits and systems, but with density, complexity, and capabilities that far exceed those found in manmade systems. [...] Obviously, silicon technology will not approach bacterial-scale integration within the foreseeable future. (pp. 848–849)

■ Engineering models for biological design

The debate over design in life is quickly coming to an end at a practical level. Evolutionists have long stated that the key difference in predictions between evolutionary and design-based models is the extent to which biological systems demonstrate suboptimal (or non-functional) bottom-up design that resembles Rube-Goldberg machines verses optimal, top-down design that resembles human engineering. The latest research has continuously affirmed the design-based predictions.

The purported examples of poor design in biology have consistently been overturned. Supposed ‘junk DNA’ is increasingly recognised as serving functions such as regulating genes, localising proteins and supporting placental development (Chuong 2018; McGrath 2021; Quinodoz et al. 2021). Biologists increasingly recognise that they need to assume life is designed to properly understand it. Biological systems are now recognised to incorporate the most advanced and effective design patterns seen in human engineering.

Before continuing, two important caveats must be mentioned. Firstly, unlike human engineering, biological design is dynamic. An organism’s design plan changes throughout its life cycle. During the development of an embryo into adulthood, different organs and systems appear, modify and disappear to sustain and direct each developmental stage to the next, and many of the features are unique to specific taxa (Ballard 1976; Kalinka & Tomancak 2012). Even after birth, the anatomy and physiology are constantly adjusting to optimise the performance of such activities as eating. For instance, fish are constantly readjusting their skull structure during growth from a juvenile into an adult to maximise their ability to ingest food and respire (Von Herbing et al. 1996).

Secondly, life represents creations far more advanced than our own. Consequently, scientists may never come close to grasping some of the underlying design concepts. Nevertheless, those aspects of life that can potentially be understood will far more closely align with engineering principles than with what any undirected process could generate.

Despite the described shifts in biological research, the philosophical resistance to openly acknowledging the fact that life is designed will likely continue for several more decades. To help accelerate biological research and mitigate materialist assumptions from retarding scientific progress, proponents of ID have initiated research projects to demonstrate the explanatory and

predictive power of design-based biological models. The developing models are founded on the principles and patterns central to human engineering, so they will be referenced as engineering models.

One set of projects addresses biological variation and adaptation. Evolutionary models predict that the total variation in a species during its and its ancestors' tenure on earth should encompass every facet of the design plan from minor details of nonessential appendages to the underlying design logic. Such limitless variation is required to explain a species' purported history of evolving from something entirely different. Evolutionary models also predict that adaptation is driven by the environment externally. In contrast, design-based models predict that variation is always limited, and adaptation is largely driven by internal mechanisms.

An organism at any given point in its life cycle is organised according to a specific design logic or architecture or body plan founded on engineering principles and motifs. The design plan includes central features that are constant and unalterable (e.g. the basic architecture of an adult fruit fly), and it includes parameters that can vary. The adjustable parameters are defined by the design logic and established during construction, and they correspond to variables (e.g. insect wing length and width) whose values are tightly bounded. The variation allows the fine-tuning of structures and systems to best function in the immediate environmental conditions.

Unlike evolutionary models, environmental happenstance plays a minimal role in setting the extreme range of survivable conditions. Whether the organism can survive within a given environment is ultimately determined by the organism's pre-set operational parameters. For instance, a lack of available food would require muscles to be maintained at a reduced size, but the muscles can only shrink by so much before a functional tolerance is exceeded. After that point, an animal could no longer move quickly enough to capture food or flee predators. It would then soon perish.

The variation in values is determined in part by the internal capacities of an individual to alter variables. For instance, an animal forced to place high stress on its limbs can often increase the strength of bones through internal adaptive mechanisms. Variation also encompasses the genetic diversity in the population, whether originating or expanding through random mutations or through NGE.

The empirical data consistently affirm engineering models. The observed variation in any species is confined to a limited set of variables such as a finch beak's thickness. No genetic variation has been identified which would allow for the basic design logic and operational parameters to change. The average thickness of beaks may increase or decrease, but such changes always remain

within a fixed range. And, more fundamentally, beaks always remain beaks. No evidence has been identified that any trait has started to transform into something fundamentally different. And no nonharmful mutation has ever been observed that would expand the genetic variation to allow such a change to occur in the future.

■ Tracking model for adaptation

One engineering model for adaptation assumes that organisms adapt to the environment using the same engineering principles seen in human tracking systems (Miller 2021d, 2021e, 2021f). More specifically, they continuously monitor the environment and track pre-specified environmental conditions. When the right conditions occur, internal mechanisms induce predetermined responses such as targeted genetic changes, physiological adjustments and or anatomical alterations.

These adaptive processes are directed by irreducibly complex systems that consistently include three components:

1. Sensors to detect pre-specified environmental conditions such as temperature.
2. Logic-based analysers that determine whether specific criteria are met such as the temperature exceeding a set point. When criteria are met, the analysers send signals to trigger the appropriate responses.
3. Effector systems and processes that generate predetermined output responses, such as growing thinner hair.

The resulting changes are targeted, rapid and often reversible. They are also predictable and repeatable. And their magnitude can range from minor alterations to dramatic transformations, but in all known cases these changes are bounded and predefined. The complexity and specificity requirements of tracking systems preclude any possibility of their having arisen through an undirected incremental process.

Over the past few decades, every facet of the tracking model has been increasingly affirmed. The strongest supportive evidence comes from studies of what has been termed NGE and phenotypic plasticity. The leading figure in NGE research has been University of Chicago geneticist James Shapiro. Notable researchers in phenotypic plasticity are developmental biologist Ralf Sommer (2020), extended synthesis proponent Sonia Sultan (Herman & Sultan 2016) and evolutionary biologist David Pfennig (2021). And the leading proponent of tracking models for adaptation in Christian circles has been Randy Guliuzza who has documented the employment of tracking systems in diverse species (Guliuzza & Gaskill 2018).

■ Natural genetic engineering

Natural genetic engineering refers to genetic alterations that are not random. Instead, they result from cells employing highly complex machinery to direct targeted DNA modifications. Shapiro summarises the processes in a 2016 review article (Shapiro 2016):

Combinatorial coding, plus the biochemical abilities cells possess to rearrange DNA molecules, constitute a powerful toolbox for adaptive genome rewriting. That is, cells possess 'Read-Write Genomes' they alter by numerous biochemical processes capable of rapidly restructuring cellular DNA molecules. Rather than viewing genome evolution as a series of accidental modifications, we can now study it as a complex biological process of active self-modification. (p. 1)

He further elaborates on the editing systems in a 2017 review article (Shapiro 2017):

Like all classes of cellular biochemistry, NGE DNA transport and restructuring functions are subject to control by regulatory circuits and respond to changing conditions [...] NGE activities typically affect multiple characters of the variant cell and organism. Consequently, major phenotypic transformations can occur in a single evolutionary episode and are not restricted to a gradual accumulation of 'numerous, successive, slight modifications'. (p. 2)

One could contest Shapiro's claims about what NGE accomplished in the past, but his general description clearly matches the engineering model's central features. The regulatory circuits that respond to environmental conditions correspond to sensors integrated with logic mechanisms. And the transport and restructuring functions correspond to specified output responses. In addition, the DNA modifications are targeted, rapid and bounded as the engineering model expects.

Natural genetic engineering has been identified in all domains of life from the simplest to the most complex. Yeast cells respond to nutrient starvation by increasing the mutation rates at specific locations referred to as mutational hot spots (Jacka et al. 2015). And the remarkable diversity in dog breeds is not the result of completely random mutations. Rather, it largely results from mutational hot spots that allow for increases in targeted genetic variation that can drive rapid adaptation. Biophysicists John Fondon and Harold Garner (2004) noted:

The high frequency and incremental effects of repeat length mutations provide molecular explanations for swift, yet topologically conservative morphological evolution [...] We hypothesize that gene-associated tandem repeats function as facilitators of evolution, providing abundant, robust variation and thus enabling extremely rapid evolution of new forms. (p. 18058)

The phrase 'topologically conservative' means that the changes are tightly bounded, just as predicted by engineering models.

Equally striking, the genomes of diverse organisms contain DNA segments known as transposable elements (TEs) that can move to new locations, allowing them to alter the activity of local genes. Specific environmental stimuli can initiate relocation to target locations (Bucher, Reinders & Mirouze 2012; Dietrich et al. 2002), and stimuli can activate the TEs, resulting in adaptive benefits. For instance, TEs modify gene regulation in maize to confer drought tolerance, alter flowering time and enable plants to grow in toxic aluminium soils (Filmer 2017; Makarevitch et al. 2015).

Investigators' philosophical bias has led them to describe TEs using terms such as selfish, parasitic or 'invasive genetic elements' (Wells & Feschotte 2020). These depictions contrast sharply with the extent to which they are often regulated (Capy et al. 2000) and targeted (Sultana et al. 2017) and with the benefits they can confer to host organisms. Such observations are forcing the use of increasingly design-based language. Miller Enguita & Leitão (2021) in a recent review article about TEs and other NGE processes commented:

[B]iological variation is the product of collective differential assessment of ambiguous environmental cues by networking intelligent cells. Such concerted action is enabled by non-random natural genomic editing in response to epigenetic impacts and environmental stresses. Random genetic activity can be either constrained or deployed as a 'harnessing of stochasticity'. Therefore, genes are cellular tools. (p. 1)

The phrase 'harnessing of stochasticity' refers to cells initiating mutations or other semi-random genetic changes in specific DNA regions to create targeted genetic variation. This increase in diversity allows a population to fine-tune specific traits to best perform in different environmental conditions.

■ Phenotypic plasticity

Phenotypic plasticity refers to an organism's ability to transform its anatomy and physiology in response to environmental stimuli. The changes do not result from genetic alterations but from internal adaptive mechanisms. Ralf Sommer enumerated these mechanisms' essential components in a 2020 review article (Sommer 2020):

[P]lasticity requires developmental reprogramming in the form of developmental switches that can incorporate environmental information. However, the associated molecular mechanisms are complicated, involving complex loci, such as *eud-1*, that function as switches and GRNs. While still early, it is likely that switch genes point to a general principle of plasticity because other examples of plasticity also involve complex switch mechanisms. (p. 7)

The 'incorporation of environmental information' tacitly implies the presence of sensors and signal transmission pathways. The switch incorporating the sensory output equates to a logic-based analyser, and the gene regulatory

network (GRN) activity corresponds to the output response. The core components perfectly match those of the engineering model for adaptation.

Phenotypic plasticity has been observed in numerous species in diverse taxa. Mice exposed to cold weather will produce offspring with shorter and fewer vertebrae (Ballinger & Nachman 2022). In a more astonishing example, gulls of the family Laridae track the sodium level in their blood with sensors in heart vessels (Sarras et al. 1985). When the level reaches a certain threshold, gulls generate a specialised gland that extracts excess sodium from the blood and excretes it through the beak. If the gull migrates to a freshwater environment, the gland disappears.

Anolis lizards raised on broad surfaces develop long legs allowing for greater sprint speeds, and lizards raised on narrow surfaces develop short legs allowing for more careful movements (Losos et al. 2000). Rove beetles respond to the presence of army ants by transforming their bodies, smell and movements to mimic the ants (Maruyama & Parker 2017). Many other examples have been identified in recent literature.

■ Model organisms confirm engineering-based models

Many model organisms have dramatically changed their anatomy and physiology as they adapted to new environments. The term model organism refers to a species that is widely studied because it is easy to breed and maintain in a laboratory setting. Key traits of interest are also accessible to the investigation such as how body shape adapts to changing environments.

The standard evolutionary story has been that traits modified gradually because of random mutations and natural selection. Such changes could potentially accumulate over extended periods resulting in large-scale evolutionary transformations. But experiments and observational studies over the past few decades are changing the narrative. Observed variation in all the most iconic model organisms originally purported to support evolutionary theory is now known to support engineering-based models.

■ Insect wings

Research on *Drosophila* fruit flies demonstrate tight constraints on wing variation. Mathematical biologist Vasyi Alba and his team analysed fruit fly wing diversity (Alba et al. 2021). The researchers employed a sophisticated analytical method to map the differences in wing structure between individual flies in a single population and between different populations. They found that the basic architecture of the wings never altered. They explained the permanence of the underlying structure as resulting from ‘global constraints within the developmental program’ that ensure similar outcomes.

In other words, developmental processes prevent wings from fundamentally altering when perturbed by mutations and environmental fluctuations. Complex mechanisms continuously compensate for such perturbations to keep the embryo's development on course. Previous research identified such corrective mechanisms as a central feature of development in complex organisms (Pezzulo & Levin 2016).

Some mutations sufficiently impact development that corrective mechanisms cannot maintain the core design architecture. Such 'macromutations' are invariably debilitating if not fatal. The negative consequences further confirm the engineering principle that any alteration of parameters beyond the predefined design constraints results in system failure. This conclusion is confirmed by the fact that no genetic variation exists in any species that would allow it to fundamentally change. Geneticist John F. McDonald called this observation a 'great Darwinian paradox' (McDonald 1983).

In addition, the researchers linked the minor variation that did exist to 'very weak mutations' and to phenotypic plasticity. Examples of the latter include a fly developing modified wings in response to changing temperature and diet. Remarkably, the team was able to map the variation from both sources to differences in the value of a single composite variable that primarily corresponds to the precise locations where wing veins intersect. Alba et al. (2021) summarise their work as follows:

Remarkably, we find that the outcomes of wing development can be statistically described by a one-dimensional (1D) linear manifold in morphological space that corresponds to a non-intuitive combination of structural variations across the wing. This dominant mode is systematically excited by variants generated by very weak mutations in signaling pathway genes as well as by thermal and dietary environmental perturbations. As such, our work provides direct empirical evidence for the presence of global constraints within the developmental program of the wing, funneling environmental inputs and genetic variation into phenotypes stretched along a single axis in morphological space. (p. 2)

These observations perfectly match the engineering-based predictions that variation only exists for predefined adjustable parameters, and the values of related variables (e.g. locations of wing vein intersections) are tightly bounded.

Other research validates the tracking model for adaptation for fruit flies by demonstrating that even the variation in the adjustable parameters did not originate from random mutations but from engineered adaptive mechanisms. The genetic variation appears to have largely resulted from NGE. Up to 80% of 'spontaneous' mutations in fruit flies are initiated by TEs (Mérel et al. 2020). Weber et al. (2005) identified several TE insertions that affect wing structure.

Multiple studies have identified other examples of phenotypic plasticity in insect wings. For instance, Bitner-Mathé and Klaczko demonstrated that fruit flies respond to changing temperature and population density by altering their wing size, shape and vein pattern (Bitner-Mathé & Klaczko 1999).

Fraimout et al. (2018) described how fruit flies respond to lower temperatures by increasing the ratio of wing size to body size. The reduction in load on the wings compensates for cold-induced lower muscle efficiency. They also described how flies alter the shape of their wings to increase the stroke force to compensate for lower beat frequency.

Gao et al. detailed an even more dramatic example in planthoppers (Gao et al. 2019). These insects track environmental cues such as temperature, food quality and population density. Depending on the conditions, they can generate two different versions (aka phenotypes or morphs). In nutrient-rich environments, nymphs develop into the 'short-winged' phenotype with reduced or missing wings, decreased flight muscle tissue and lower energy metabolism. This form is more efficient in a local environment with abundant food. More importantly, this phenotype reproduces earlier and produces more eggs, which benefit the population in such favourable conditions.

The 'long-winged' phenotype is generated if the environmental conditions deteriorate because of reduced food supply or overpopulation. This version has a well-developed flight apparatus with longer wings, larger flight muscles and an enhanced energy metabolism. It can escape the unfavourable environment and colonise new locales.

The switching mechanism between the phenotypes is exceedingly intricate. Environmental sensors feed readings to the IIS signaling pathways that are commonly employed in complex organisms to control tissue growth and body size. The IIS signaling component triggers a host of other mechanisms that direct hundreds of genes including the production of alternative versions of proteins. The ingenuity and complexity behind switching mechanisms in even the simplest of animals (Sieriebriennikov et al. 2018), and the other referenced mechanisms not only highlight the extraordinary design behind biological adaptation, but they also reinforce the need for engineering models, tools and language to broaden and expand our understanding of living systems.

■ Cichlids

The dramatic variation in cichlid fish has become a poster child for the creative power of evolutionary processes, but research over the past 15 years has shown the opposite. Cooper et al. demonstrated that variation in cichlid feeding is tightly constrained (Cooper et al. 2010). The investigators used a principal component analysis (PCA) to quantify differences in feeding structures for cichlid populations in Lakes Tanganyika, Malawi and Victoria. This technique creates composite variables that capture how the value of one variable (e.g. head width) correlates with another variable (e.g. jaw length) or variables for each fish. The analysis revealed that variation in all populations largely resided along the same two composite variables (also known as principal component axes).

Kara Feilich performed a PCA on cichlid body and fin structures, and she also found that the same variation occurred repeatedly in separate populations (Feilich 2016):

Body shape disparity among the cichlid fishes has been studied extensively, repeatedly demonstrating common axes of diversification across many lineages, including the tropheines (Wanek and Sturmbauer 2015), geophagines (Astudillo-Clavijo et al. 2015), and others (Clabaut et al. 2007; Muschick et al. 2012). (p. 2260)

Feilich noted that changes in traits were highly correlated to optimise hydrodynamics and manoeuvring. Of key importance, the underlying design architecture never changed. Only the flexible parameters were adjusted to fine-tuned individuals in a population to best perform in its immediate environmental conditions.

Other studies identified phenotypic plasticity in multiple traits. Mazzarella et al. raised cichlids in water with different salinity levels. Greater salinity resulted in adult fish displaying shallower bodies and longer jaws (Mazzarella et al. 2015). Härer et al. (2019) exposed Midas cichlids to light of different frequencies. In response to a change in frequency, the cichlids switched the expression of cone opsin genes crucial for colour vision in only a few days.

Parsons et al. fed two separate groups of young juvenile fish different diets (Parsons et al. 2016). The two groups developed into adults with distinctly different head-jaw structures that were tailored to forage for the available food most effectively. Navon et al. reported on a similar experiment that confirmed the Parsons et al. results (Navon et al. 2021). The investigators also demonstrated diet-induced adaptive changes to body shape and fin-ray number. The observed dissimilarity between the two diet groups mimicked the differences between distinct cichlid species in the wild.

Researchers increasingly recognise that the most significant variation in cichlid fish results not from random mutations but from internal adaptive mechanisms. Parsons et al. (2016) noted:

[7]here is an emerging view that additive genetic variation accounts for a relatively small percentage of phenotypic variation and rather it's the context in which traits develop that determines their final form (Hallgrimsson et al. 2014; Hendrikse et al. 2007; Jamniczky et al. 2010; Pfennig et al. 2010). (p. 6021)

Other investigators discovered that some of the key genetic variation that did exist resulted from genetic alterations that point to NGE. Carleton et al. demonstrated that some of the most important variation resulted from TEs that were targeted to the genes underlying vision (Carleton et al. 2020).

■ Sticklebacks

As with cichlids, all studies on stickleback fish diversity demonstrate that variation is tightly constrained. Hohenlohe et al. documented how the same

variation in traits and genetics appears repeatedly in separate populations (Hohenlohe et al. 2010):

Genomic regions exhibiting signatures of both balancing and divergent selection were remarkably consistent across multiple, independently derived populations, indicating that replicate parallel phenotypic evolution in stickleback may be occurring through extensive, parallel genetic evolution at a genome-wide scale. (p. 1)

Miller et al. came to the same conclusion after analysing the genetic differences between populations that inhabited lakes in the presence and absence of prickly sculpin, a fish that is a stickleback predator (Miller, Roesti & Schluter 2019). Sticklebacks that interacted with prickly sculpin rapidly acquired similar alterations to hundreds of the same genes. The two studies demonstrate that sticklebacks adapt genetically to environmental changes predictably and rapidly.

These observations suggest that NGE might be driving targeted genetic alterations. Another study has identified NGE more explicitly. Ishikawa et al. discovered that multiple stickleback species duplicated the *Fads2* enzyme allowing the species to better synthesise the essential fatty acid DHA (Ishikawa et al. 2019). This enhanced ability allowed them to colonise DHA-deficient freshwater environments. The authors suggest that the duplications were facilitated by NGE, possibly the relocation of TEs.

Other investigators identified examples of phenotypic plasticity. McCairns and Bernatchez discovered that sticklebacks inhabiting freshwater and saltwater zones of a large estuary measure the salinity of the water (McCairns & Bernatchez 2010). They use this information to optimally regulate the expression of genes controlling the transport of salt ions, so the fish can quickly adapt to salinity changes.

Baker et al. demonstrated that stickleback females track internal physiological information such as lipid supply and liver glycogen level and environmental cues such as availability of food and population density (Baker et al. 2015). Different cues initiate adjustments to such reproductive parameters as a time of breeding, egg size and clutch or brood size. The alterations improve the likelihood for the population's continued survival. The researchers also discovered that individual fish coordinate the fine-tuning of multiple traits to ensure optimal reproductive success.

As a final example, Tibblin et al. (2020) raised sticklebacks in an aquarium with different colour backgrounds. The investigators also mimicked the presence of predators by chasing fish with a dip-net and introducing chemical cues mimicking the presence of Pike, which is a natural predator. Both colour and predatorial stimuli triggered changes in the dorsal colouration that assisted the fish in avoiding detection.

■ Cavefish

Cavefish also demonstrate phenotypic plasticity for multiple traits. Rohner et al. raised *A. mexicanus* embryos in water with low conductivity mimicking cave conditions (Rohner et al. 2013). The embryos developed into adults with significantly smaller eyes. Corral and Aguirre raised *A. mexicanus* in different temperatures and different levels of water turbulence (Corral & Aguirre 2019). The variant conditions resulted in adult fish differing in vertebral number and body shape. For instance, fish raised in more turbulent water displayed more streamlined bodies and extended dorsal and anal fin bases that improved their mobility in that environmental condition. Bilandžija et al. raised the same species in darkness, and the fish developed many cave-related traits such as resistance to starvation and altered metabolism and hormone levels (Bilandžija et al. 2020). Future studies will likely demonstrate more examples of phenotypic plasticity and NGE.

■ The collapse of evolutionary icons

Cichlid, stickleback and cavefish represent three of the most iconic examples of adaptation that biologists present as evidence for the plausibility of evolutionary processes driving large-scale transformations. Yet research over recent decades demonstrates that evolutionary and adaptive processes are constrained in these fish, as in all complex organisms, to only minor alterations to existing traits or to the loss or duplication of an existing structure.

Some random mutations do occur that confer benefits, but nearly all of them either degrade genes or cause modifications so trivial that they could never accumulate to fundamentally alter any trait. This conclusion was confirmed by an extensive survey of the technical literature conducted by Michael Behe in his book *Darwin Devolves: The New Science About DNA That Challenges Evolution* (Behe 2019).

Behe specifically addresses studies of the variation in cichlid genes. One investigation identified amino acid changes in a gene affecting colour patterning. The mutant genes appeared to lose certain abilities, so the mutations degraded the original gene. Another study identified a mutation in a gene that conferred slightly improved light sensitivity at certain depths (Nagai et al. 2011). In this case, only a single amino acid in one protein changed. In contrast, nearly all the most significant adaptation results from carefully controlled processes that leave little to blind chance.

■ Engineering models and the pattern of life

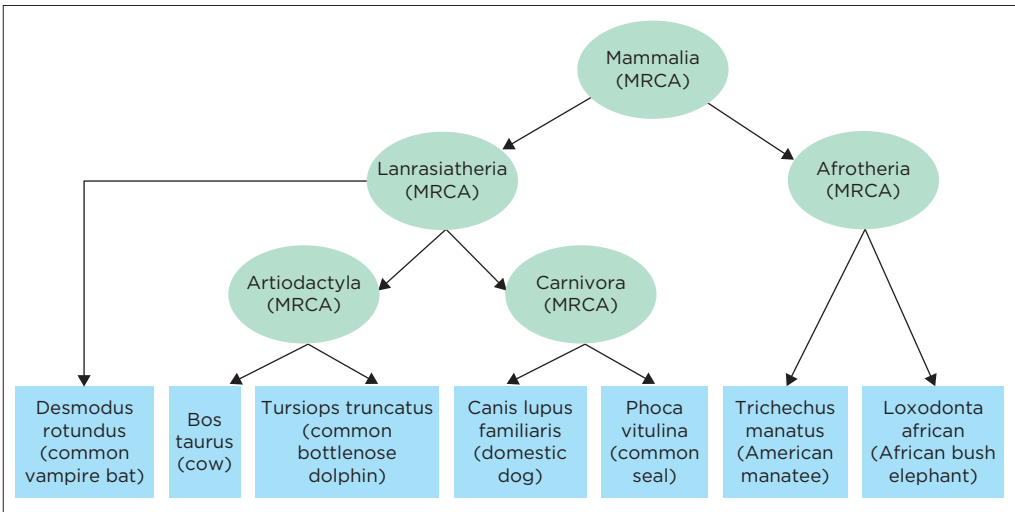
Engineering-based models also better explain the pattern of similarities and differences between species in the higher taxonomic groups (e.g. phyla, classes and orders) than the theory of common ancestry (Miller 2018, 2021b).

Computer scientist Winston Ewert demonstrated that the distribution of the same gene families in diverse species far better fits what he refers to as a dependency graph model than the common ancestry model (Ewert 2018). His central conclusion is that similarities in life represent modules that were implemented in diverse species to achieve similar goals. This general conclusion has been validated by multiple lines of research over the past few decades.

■ The collapse of the tree of life⁵⁷

One of the central pillars of the standard evolutionary model is the belief that all living species evolved from a common ancestor through a gradually unfolding TOL. As a result, the theory predicts that the pattern of similarities and differences in species today should fit into a tree-like pattern or nested hierarchy where branching points correspond to the appearance of new traits. For instance, all mammals share certain features, such as producing milk, since their most recent common ancestor first evolved them. Those traits carried through each evolving branch of the mammal tree (Figure 7.2). The claim that all species fit into the TOL has been presented to the public as one of the strongest pieces of evidence for the belief that all of life evolved through undirected processes.

What the public is rarely told is that the pattern of physical traits and molecular sequences in species does not fit into a consistent evolutionary



Source: Ewert (2018).

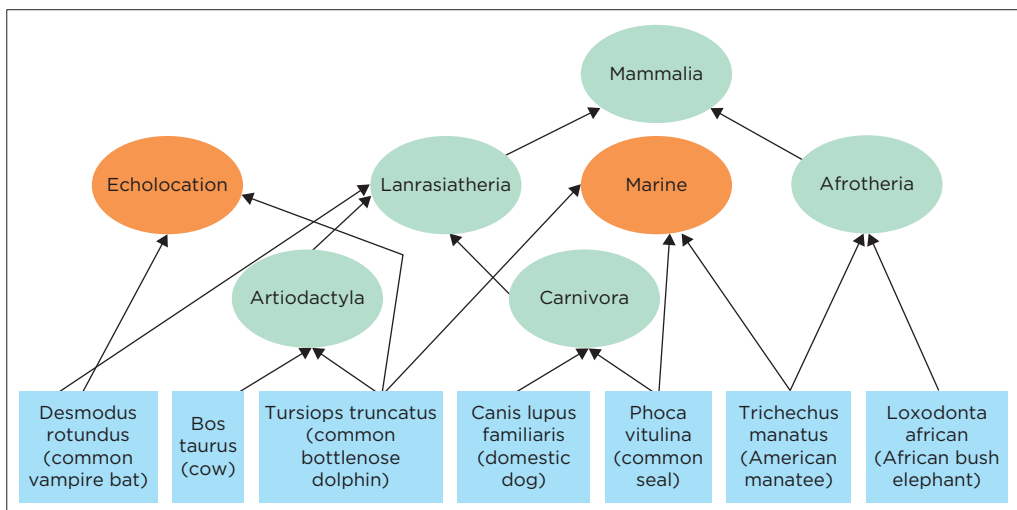
FIGURE 7.2: A subset of the mammalian tree of life. Rectangles are extant species, and ellipses are postulated ancestral species, the most recent common ancestor of each taxonomic category.

57. This section represents a reworking of excerpts from Miller (2018).

tree (Figure 7.3). As an iconic example, bats and dolphins both possess echolocation, which is undergirded by the same modifications at nearly 200 locations in their genomes (Parker et al. 2013). The species are not believed to be closely related, so the similarities must have originated independently in direct conflict with the prediction of common ancestry. Such conflicts are frequent and widespread in nearly all taxa.

Zoologists Klassen, Mooi and Locke (1991) summarised across numerous studies the percentages of traits that fit consistently with the best-constructed evolutionary trees. The researchers plotted the percentages (also known as consistency indexes) on the same graph as those derived from randomly generated data, and the values were then adjusted to remove the effect of random noise. The results indicate that most of the data do not fit the prediction of common ancestry.

More recent research on the rescaled consistency indexes (RCI) of numerous animal groups confirms this conclusion. An RCI of 1 indicates that all the data fit consistently with an evolutionary tree. An RCI of 0 indicates that the appearance of similarities is completely random. Data of the traits in arthropods yielded an RCI of 0.39 (Wolfe 2017). The RCI for therapsids (purported ancestors to mammals) was 0.42 (Huttenlocker 2009). The RCI for primates and their purported closest relatives was 0.29 (Bloch et al. 2007). And the RCI for Cetaceans (e.g. whales and dolphins) and their purported closest relatives was 0.24 (Thewissen et al. 2001). Evolutionists cite these groups as representing some of the strongest evidence for common ancestry, but their low RCI values demonstrate the opposite.



Source: Ewert (2018).

FIGURE 7.3: A possible subset of the mammalian dependency graph of life. Rectangles are species, and ellipses are postulated modules. The orange ellipses are modules postulated in addition to the standard taxonomic modules.

Evolutionists historically predicted that RCI values would fall far closer to 1 than to 0, which they clearly do not. The low values demonstrate that the assumption that similarities in traits reliably imply common ancestry is false. The problem has become so acute for microorganisms that microbiologists Merhej and Raoult described attempts to identify the TOL in the most disparaging of terms (Merhej & Raoult 2012):

None of the seven points laid out in the introduction to this manuscript can be permanently retained, as established by Darwin's theory, which was at the time a fight against the creationists. This theory cannot be upheld in its entirety. Recent advances from genomics refute the ideas of gradualism, exclusive vertical inheritance, evolution selecting the fittest, a common ancestor and the TOL. Indeed, there may not be any two genes that have the same evolutionary tree. (p. 36)

These disappointing results have forced evolutionists to propose several mechanisms to explain the recurrent inconsistencies. Examples include lateral gene transfer (LGT), differential gene loss, incomplete lineage sorting and convergent evolution. Lateral gene transfer refers to genes passing from one organism to another. This process could theoretically explain how the same genes appear in unrelated species, but the plausibility of widespread LGT in complex organisms has been seriously questioned (Martin 2017). Incomplete lineage sorting and gene loss cannot explain complex traits appearing in distantly related organisms. The claim that complex adaptations can evolve independently multiple times (i.e. convergent evolution) collapses on close examination because of the implausibility of their appearing through undirected processes even once.

For instance, eyes with lenses are believed to have evolved independently multiple times, but all evolutionary scenarios face insurmountable barriers. The first mutations in the origin of a lens would allocate tissue in front of the photoreceptors. The problem is that undifferentiated tissue would degrade light reception, so the first mutations would quickly disappear. The lens would not become beneficial until a complex developmental process coupled to a new GRN emerged (Ogino et al. 2012). Yet the available time based on the fossil record is insufficient for even the tiniest fraction of required new genetic information to arise (Hössjer et al. 2021).

■ Engineering modules

In contrast, engineering-based models satisfactorily explain the pattern of similarities throughout life. Design architectures often fall into a hierarchical pattern. All transportation vehicles have certain common features such as allocated space for cargo and or passengers, propulsion system and steering. Cars possess all these features plus such components as wheels, breaks, coolants, lubricants and axles. Toyota Camry models possess all these features plus additional specialised components. The similarities in transportation

vehicles would likely fit into a constructed tree at least as well as different groups of species.

While many features in human products fit into a hierarchical tree-like pattern, many break that pattern. A police car and an airplane both have two-way radios while two-way radios are absent in most other cars. In addition, the same circuitry is implemented in a wide variety of vehicles to meet similar goals. This pattern reflects how engineers often create modules that can be used in diverse contexts. The modules must be designed with the explicit intent of operating in different products, and the products that use the modules must be designed to properly incorporate them into their operations. The same pattern and constraints are observed in life.

Many of the same traits are implemented in diverse creatures to perform similar functions. Different versions of eyes (e.g. compound and camera-like) appear independently multiple times to allow optimal vision in each species' particular environment. And very similar neural and developmental modules appear independently in unrelated species. Neuroscientists Sanes and Zipursky identified in both fly and human visual systems remarkably similar design motifs in the retinas, neural circuits in the brains and genetic control mechanisms in development (Sanes & Zipursky 2010) even though the different eyes are believed to have evolved independently (Piatigorsky 2008).

Similarly, mathematical biologists Robyn Araujo and Lance Liotta (2018) demonstrated that all biological networks that perform 'robust perfect adaptation' (RPA) can be broken down into two distinct classes of modules. The researchers define RPA as (Araujo & Liotta 2018):

[T]he ability of a system to generate an output that returns to a fixed reference level (its 'set point') following a persistent change in input stimulus, with no need for tuning of system parameters. (p. 2)

The same modules appear in systems as diverse as signal transmission, gene regulation, protein interaction networks, sensory systems and developmental regulation. They are often combined hierarchically to perform more complex functions.

Of key importance, the base modules operate within exacting constraints:

1. Their constituent components must interconnect according to specific blueprints.
2. They cannot be subdivided, so they comprise a set of components that are irreducibly complex.
3. Their operations must solve a local adaptation equation, so reactions or other processes must operate within tight bounds.
4. The integration of the modules into larger systems must also meet rigid mathematical criteria.

These constraints imply that the modules' origin and implementation could not have occurred incrementally through an undirected process as their construction and integration requires coordination, foresight and goal-direction. Szallasi, Stelling & Periwai (2007) assume evolution must be true as a faith commitment, but they still comment:

[T]he concept of 'modular design' is borrowed from human engineering and therefore has an essentially forward looking, goal-oriented nature. Complex engines and networks are constructed from modules while the final overall behavior of the system is kept in mind. (p. 44)

The pattern of nature also corresponds to engineering principles in the distribution of species diversity. In human engineering, many versions of a particular design are often created. Each year, Toyota manufactures the same car model with different sets of modules such as cameras and tracking systems. But few products would fall outside of the larger categories. Vehicles have rarely been designed that would fall part-way between a submarine and an airplane.

Similarly, many different species exist that represent different versions of the same theme. Several ape-like and human-like creatures have been identified that share various similarities. But the similarities do not fit within a consistent evolutionary tree (Collard & Wood 2000). And clearly intermediate creatures between primates and other orders of mammals have never been identified (Gingerich 2006). In summary, the pattern of nature does not point to an undirected evolutionary process but ID.

■ Case studies on engineering models

Two case studies will further illustrate engineering models' explanatory and predictive power. The first focuses on minimally complex cells and the origin of life. The second describes an engineering-based analysis of the bacterial flagellum.

■ Minimally complex cell

The two chapters by James M. Tour and Fuz Rana detail the implausibility of a cell ever originating through undirected natural processes. I have previously explained the insurmountable thermodynamic challenges to a chemical system spontaneously coalescing into an autonomous cell (Miller 2020a, 2020b). These analyses are critical to the design argument. The only explanations for life, or any artifact, are natural processes, chance, and design (Dembski 2006). The fact that natural processes and or chance cannot explain life's origin in itself supports the design inference. The addition of the following positive evidence makes the identification of the design conclusive.

Engineering analyses elucidate in even the simplest of cells much of the underlying architecture and the clear top-down design logic. Top-down logic requires levels of foresight, coordination and goal-direction that only intelligent agents can implement. Such evidence has forced even scientists who philosophically reject design to unconsciously recognise in life's origin its necessity.

Systems engineers have analysed the minimal requirements for self-replication in such systems as lunar factories (Freitas 1980) and robots (Moses & Chirikjian 2020). The research was guided by the theoretical work of mathematician John Von Neumann (1966). In parallel, several studies have attempted to identify the minimal components of an autonomous cell (Xavier, Patil & Rocha 2014). The confluence of these two streams of research demonstrates that a minimally complex autonomous cell must possess at least the following components:

1. Machinery for energy production and delivery.
2. Information repositories and processors.
3. Selective gateways with active transport.
4. Sensors coupled to signal transduction pathways and signal processing.
5. Actuators that implement instructions.
6. Manufacturing and auto-assembly processes.
7. Automated repair machinery.
8. Error-correction systems.
9. Waste disposal and recycling mechanisms.
10. Control systems capable of global coordination.

The theoretical engineering analyses match not only many of the components of minimally complex cells but also their interrelationships and underlying design logic.

This correspondence demonstrates that cells display top-down design where an overarching design architecture determines the higher-level systems and their interdependencies. Each system also operates according to its own design pattern (e.g. control feedback loop) that determines the lowest-level processes. The design patterns are disassociated from and transcend the physics and chemistry of the underlying molecules, so little that occurs in a cell would happen without external management.

Leading origin-of-life researchers Elbert Branscomb and Michael Russell explain how cells must employ molecular machines to manage such processes as chemical reactions (Branscomb & Russell 2018a):

But even those of life's molecular transformations that do run downhill have to be taken out of chemistry's hands and 'managed' by a dedicated macromolecular machine - in order to impose conditionally manipulable control over reaction rates and to exclude undesirable reactions, both as to reactants and products. On its own, chemistry is far too indiscriminate and uncontrollable [...]. (p. 1)

They even acknowledge that everything must conform to an ‘elaborate organisational design’.

The researchers state that molecular machines are not only required in life today, but they were also essential at its emergence (Branscomb & Russell 2018b). Remarkably, they also acknowledge that even the most basic elements of life could never have arisen through natural chemical processes:

We claim in particular that it is untenable to hold that life-relevant biochemistry could have emerged in the chemical chaos produced by mass-action chemistry and chemically nonspecific ‘energy’ inputs, and only later have evolved its dauntingly specific mechanisms (as a part of evolving all the rest of life’s features). (p. 4)

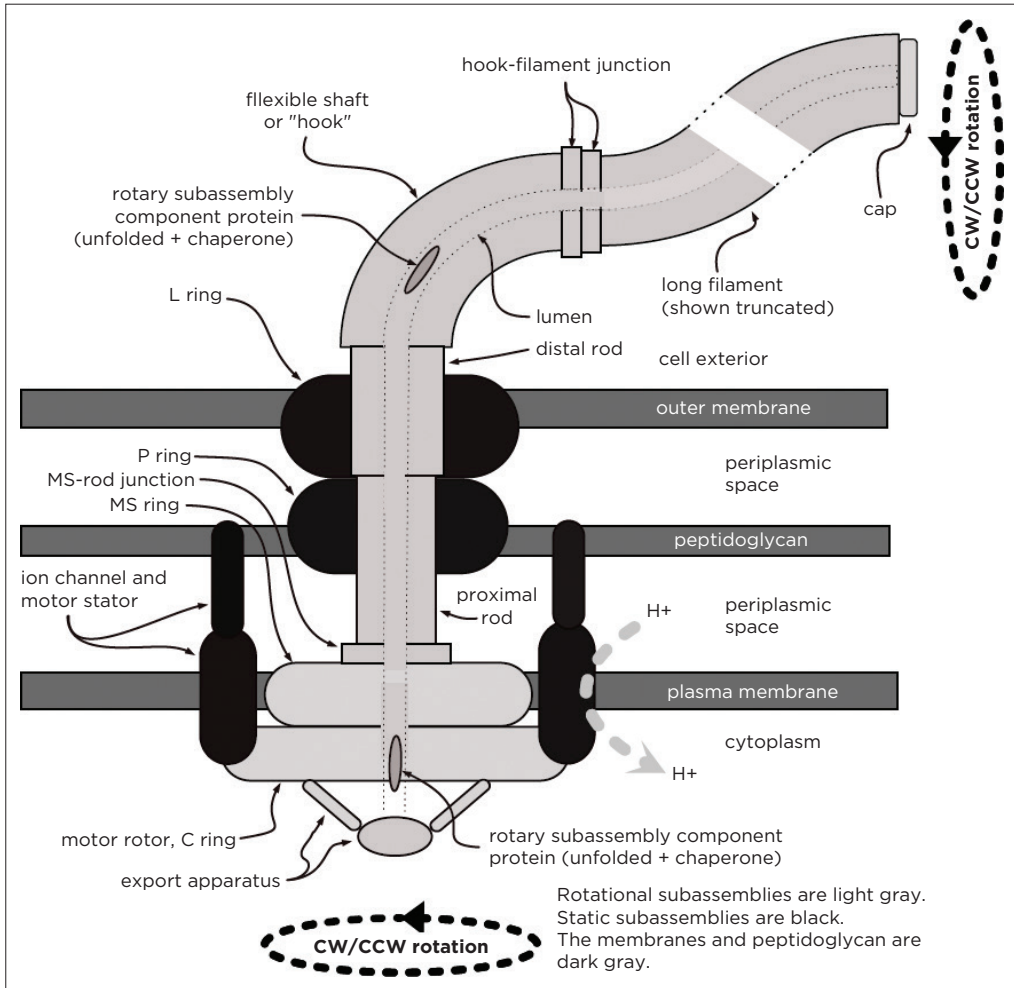
They respond to this seemingly insurmountable barrier by appealing to natural selection as a guiding hand to bring every system into being and properly interconnect them. The obvious problem is that natural selection could not have commenced before an autonomous self-replicating entity appeared. The authors in effect tacitly acknowledge that life is the product of ID, but they use the language of natural selection to smuggle intelligent agency into their theory without giving due credit to the designer.

■ Bacterial flagellum

The bacterial flagellum functions as a rotary propulsion system in bacteria (Figure 7.4) and showcases the predictive power of engineering models (Miller 2022). The system employs many stunningly complex subsystems. The assembly operation is directed by a genetic network (Fitzgerald, Bonocora & Wade 2014) that ensures the manufacture of the right proteins in the right quantities at just the right time. The proteins that compose the propeller are transported across the cell membrane through a transport gate that only allows the correct ones through at the correct time by employing protein signal sequences (Evans, Hughes & Fraser 2014).

Other proteins assemble the hook and propeller (Kim et al. 2017; Ohnishi et al. 1994). Another protein acts as a ruler that coordinates with the gate to ensure that the flagellar hook extends to the correct length (Waters, O’Toole & Ryan 2007). A navigational control system controls the rotation of the rotary motor (Hamadeh et al. 2011).

Computer scientist and engineer Waldean Schulz approached the design of the flagellum with a method that could be described as ground-breaking. He first outlined what he expected would be the most efficient architecture for a nanotechnology rotary propulsion system (Schulz 2021a). He started with the highest level in the organisational hierarchy and mapped the overarching design of the core processes, including manufacturing, assembly, energy production, torque generation, environmental tracking and directional control. He graphed the expected minimal components and their functional relationships. He also anticipated design requirements and constraints.

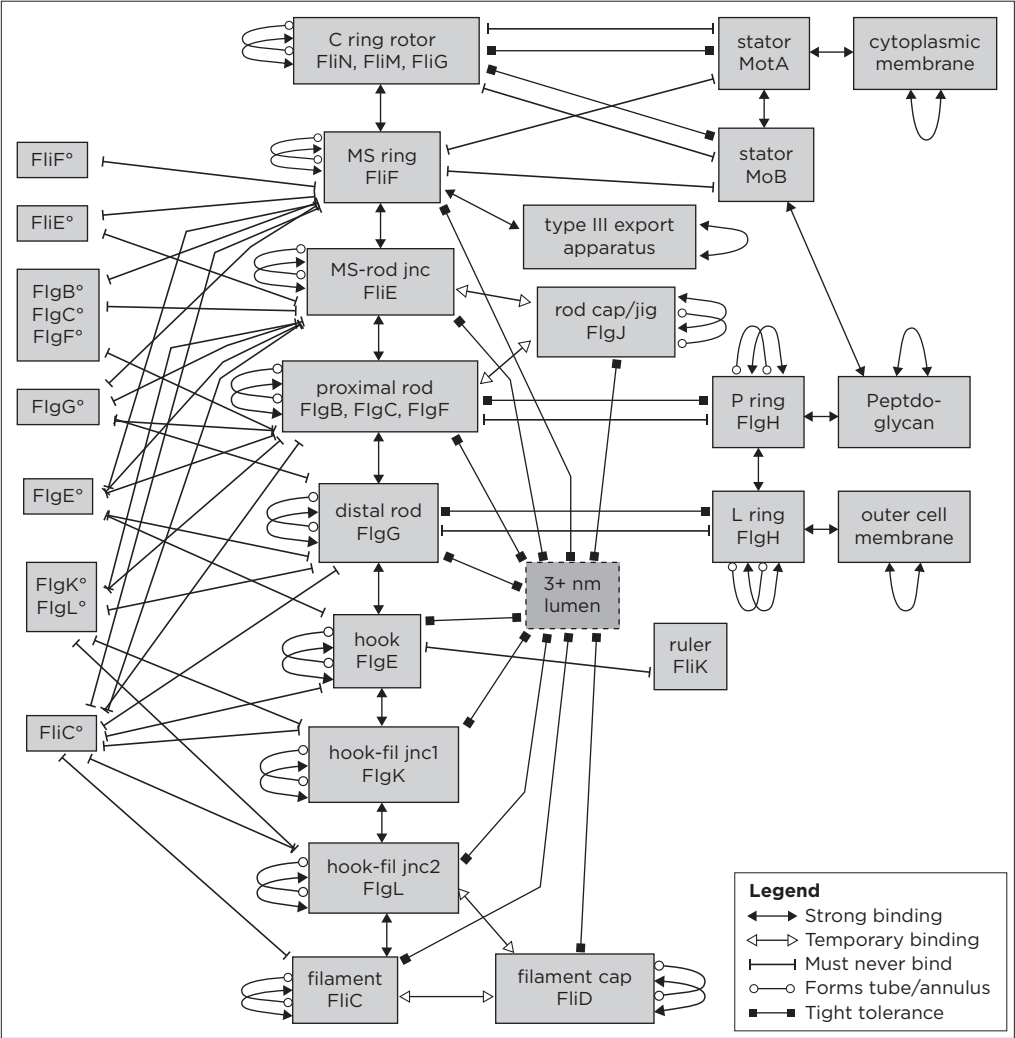


Source: Schulz (2021a).

FIGURE 7.4: Structure of an archetypical flagellum.

He then reviewed the literature on actual flagellar operations starting with the processes at the bottom organisational level (Schulz 2021b). Finally, he compared the top-down and bottom-up analyses (Schulz 2021c). Schulz's expectations for the design architecture, interrelationships and constraints match the actual operations to a remarkable extent.

In addition, he identified numerous tight constraints that must be met for the flagellum to function at an efficiency that would provide any benefit to the cell. The constraints include over 80 requirements for the interactions between the individual proteins (see Figure 7.5). For instance, some proteins must bind together permanently, others temporarily, others must never bind. If any one of these criteria is not met, the flagellum would provide no benefit to the cell. It would instead disadvantage the cell by wasting resources.



Source: Schulz (2021b).

FIGURE 7.5: Binding and geometrical properties of the proteins.

Schulz’s analysis demonstrates that the flagellum was engineered around a clear overarching design logic where every system and component was optimally designed to integrate with multiple other systems and components in symphonic harmony. Each of these systems is required for proper operations, and they must coordinate with each other with extremely high efficiency. The number of essential components, interrelationships, requirements and constraints in the flagellum proves that the system must have originated at once through the act of an intelligent agent. Schulz’s work is simply a foretaste

of the long-term clarity and insights engineering-based models will bring to the study of living systems.

■ Future research

Future researchers can follow Schulz's example by identifying the engineering logic underlying other living systems. They can then also anticipate the core components, their interrelationships and the operational constraints by comparing the systems to comparable human creations. This task is only possible because life does not employ design patterns and processes that are analogous to those used in human engineering. But they are the same patterns and processes.

Life does not employ systems analogous to information-processing, but they are synonymous with information-processing. And cells do not replicate DNA using systems analogous to error correction, but they perform error correction. This recognition will avoid the misdirection that often occurs from the false assumptions attached to scientific materialism (e.g. suboptimal bottom-up design), and it will accelerate progress in biological research.

■ Conclusion

The field of biology has just begun to enter the next great scientific revolution where the philosophical pendulum has started to swing away from scientific materialism and back towards the design-based philosophical traditions of Plato and Aristotle. The sheer weight of the evidence has forced this shift upon the scientific community. In particular, the central predictions of evolutionary theory have consistently failed while the predictions of ID have been increasingly confirmed. This paradigm shift is clearest where scientists have most incorporated engineering principles and insights into their biological research.

Incorporation of design concepts: Biologists have only now begun to tap into engineering models' explanatory and predictive power. Schulz's application of engineering principles to analysing the flagellum allowed him to anticipate much of the rotary system's structure and operations. As his top-down integrated with bottom-up approach is applied to other systems, investigators will increasingly recognise that the only viable framework for understanding life is starting from the assumption of design. This trend will only accelerate as discoveries continue to demonstrate the top-down nature of biological design, the optimality of living systems and the tight constraints intrinsic to biological systems.

Many systems biologists have already come to recognise that they must incorporate the core ID concepts into their analyses, albeit using different

language, to advance their research. The heuristic of ID is simply a more general rubric for the application of engineering principles to the study of life. More specifically, Behe's concept of irreducible complexity is implicit in the tenet of holism as described earlier. William Dembski's formulation of specified complexity is a more general category for biologists' understanding of design modules.

The argument for design: This last example is of critical importance. Specified complexity entails a pattern that is not the product of a natural process, extremely rare and matches a special externally defined pattern (Dembski 2006). Patterns that demonstrate specified complexity beyond a certain threshold are always the product of design. An example would be Scrabble letters arranged to spell out an intelligible paragraph. Life demonstrates many examples of specified complexity; therefore, Dembski has argued that the conclusion that life is designed is nearly certain.

Yet evolutionists challenged Dembski by countering that one cannot conclusively demonstrate that a pattern in biology is in any way special. They also asserted that a biological entity might appear highly improbable to have originated by chance, but many other biological solutions might also have played the needed role. The probability of finding one solution out of the multitude might be tractable.

The observation that life employs the same design motifs seen in human engineering as modules overturns these objections. A design motif is a highly improbable, special pattern. Relatively few exist in biology that solve engineering tasks such as RPA. In addition, the fact that the same motifs used in human engineering are implemented repeatedly in life demonstrates that multiple other options do not exist that life could have stumbled across by chance.

The theological implications: This conclusion and the corresponding paradigm shift in biology entail substantial theological implications. Many Christians have attempted to reconcile evolutionary theory with Christian theology, but their efforts have often resulted in jettisoning or distorting historic Christian doctrines and twisting biblical texts. Or they have portrayed evolutionary theory in terms that directly conflict with how the theory is generally understood and taught (Moreland et al. 2017). Now, the scientific evidence is so decisively in favour of design that Christians no longer need to waste time in the futile task of accommodating Christian theology to secular creation narratives.

In addition, the design patterns seen in life demonstrate that God created the world so that specific engineering motifs work effectively to achieve such goals as vision or locomotion. This realisation explains the striking

similarities between human creations and living systems, and it explains how Schulz was able to predict so many details of the flagellum's structure and operations.

Christians can now speak with complete confidence about the clear evidence of God's hand in life. Many atheists and agnostics who hear the truth will continue to suppress it, but for others the evidence will break their minds free from secular philosophies. They will then gain the freedom to pursue a relationship with their Creator.

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Chapter 7

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Chapter 8

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